

1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

There are various applications which use different techniques to secure the application.
Five of the real-world applications are listed below.

Application	Security Feature	Security Mechanism	Protocol/ Technology	How it provides security
Razorpay	Confidentiality	Card tokenization (no raw card data stored), encrypted transmission	RBI Card-on-file Tokenization(COFT), TLS, AES-128	By law Razorpay is not allowed to store card data, but rather it stores a scrambled token combined with encryption during transmission
Netflix	Authorization	Content encryption and license based playback control	Digital Right Management(DRM) , (widevine, FairPlay)	The video file by default is encrypted and locked, the device with the subscription, requests a "key" to unlock and play.
BookMy Show	Availability	Rate limiting, auto-scaled load balancing across regional clusters	AWS API Gateway/ Nginx, Redis, Elastic Load Balancers	During huge traffic spikes example popular concert the system limits how many request a single user can make and it spreads the incoming traffic across multiple server.
Icloud service	Integrity	File checksums for de-duplication and tamper detection	Cryptographic hashing	Every file the user stored gets a unique digital footprint, if a file gets corrupted while syncing , the fingerprint won't mach anymore so that iCloud can detect the problem.
X(Twitter)	Authentication	Two-factor authentication via authenticator apps, security keys, pass keys or OTP	2FA (TOTP), Passkeys	Even if the password is compromised the attacker will not be able to log in without a second code from an authenticator app, pass key or top

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

The security controls for all the applications can be classified into 3 major types: Confidentiality, Integrity and Availability.

Below are listed five real-world applications.

Applicaton	Security control	Protocols	Prevention	Detection	Recover
RazorPay	Confidentiality	Card Tokenization, TLS, AES-128, RBI COFT	Tokenization prevents attackers from accessing real data	TLS/Certificate error flags interception attempts	Remove and issue compromised token
Netlix	Authorization	DRM (Widevine, FairPlay), license-based playback	Content stays encrypted, until the appropriate key is generated	License server, logs abnormal key request	Revoke the device license
Bookmyshow	Availability	Rate limiting, Load balancing	Limits request per user, blocks overload traffic and restrict bots access	Traffic monitoring flags abnormal traffic	Auto scaling spins up server to restore services
iCloud	Integrity	Cryptographic hashing, file checksums	Hash verification rejects tampered files	Checksum mismatch during synchronization flags error	Resynchronization or restore the last version
X(Twitter)	Authentication	Two-factor authentication via authenticator apps, security keys, pass keys or OTP	Compromised password is not enough for attacker to gain access	Failed two factor authentication alerts user about the unauthorized access	Reset password and creating a new session with updated passkey